

THE OPPORTUNITY BEYOND THE FAB

Free Edition - the thesis, the framework, and the top ten of 100 opportunities

Techadyant Labs

Table of Contents

The Opportunity Beyond the Fab — Free Edition

India has committed roughly ₹1.6 lakh crore to semiconductor manufacturing — thirteen projects across seven states, with the first facilities now operational. The national conversation treats this as the story. It is not. It is the opening of a much larger one. This is not an argument against the fab: the fab is the anchor that creates the surrounding demand, and many of the opportunities mapped here exist precisely because the fab investment is happening. The argument is about where the value is *captured* — around the fab, not instead of it.

A fab is the visible peak of a ten-layer industrial pyramid, and the layers around it — the materials and precision components beneath, the design tools and software beside, the applied AI, security and services above — are where most of the capturable economic value sits, where the capital requirements are a fraction of a fab's, and where India's existing industrial and talent base can actually compete. The full report identifies, scores and ranks **one hundred startup and MSME opportunities** created by India's technology-sovereignty push. This free edition gives you the thesis, the framework, and the top ten.

The framework: the India Technology Sovereignty Stack

The organising structure is the India Technology Sovereignty Stack — ten layers from raw materials to research. India's position across it is uneven in a way that defines the opportunity. India is **strong** at the downstream end (electronics manufacturing, where ninety-nine per cent of phones sold are now locally made; design services, where India performs roughly a fifth of the world's chip-design work). It is **emerging** across the middle — manufacturing, software, AI, engineering tools, security. And it is **missing** at the upstream end — materials, chemicals and capital equipment, almost entirely imported, where roughly ninety to ninety-five per cent of demand is met from abroad (NITI Aayog, 2026).

Each position implies a different kind of opportunity: convert capability to ownership where India is strong; climb from participation to value capture where it is emerging; build behind a sovereignty mandate where it is missing.

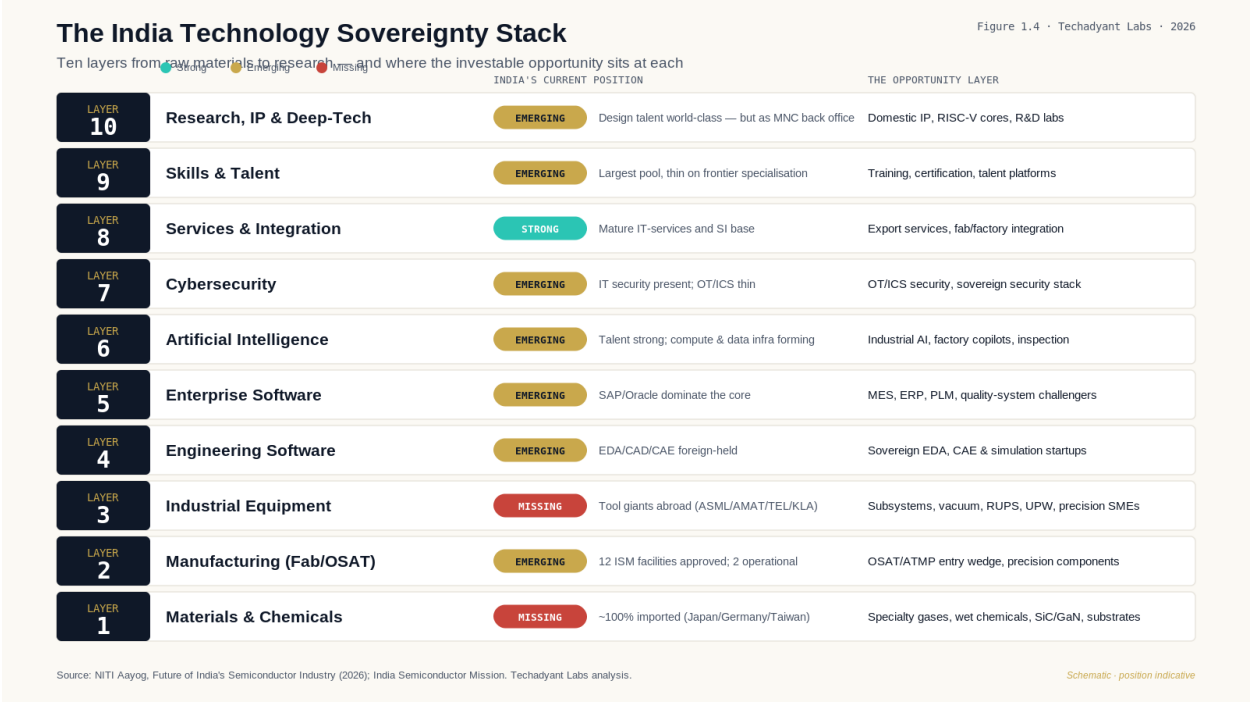


Figure 1 — The India Technology Sovereignty Stack: ten layers from materials to research.

How the 100 opportunities are scored

Every opportunity is scored on a single, reproducible system of five proprietary indices. Three diagnostics — the Technology Sovereignty Value Index (how strategic), the India Readiness Rating (can India compete today), and the Go-to-Market Ease Matrix (how hard to commercialise) — feed one headline Technology Opportunity & Market Index (TOMI). A portfolio engine, the Strategic Opportunity Engine, then sorts the hundred into four classes: 21 Strategic Quick Wins, 58 National Bets, 15 Cash Engines, and 6 to watch.

The Top-100 opportunity landscape

Each bubble = one opportunity. x = market attractiveness, y = go-to-market ease, colour = portfolio class.

Figure 0.2 · Techadyant Labs · 2026

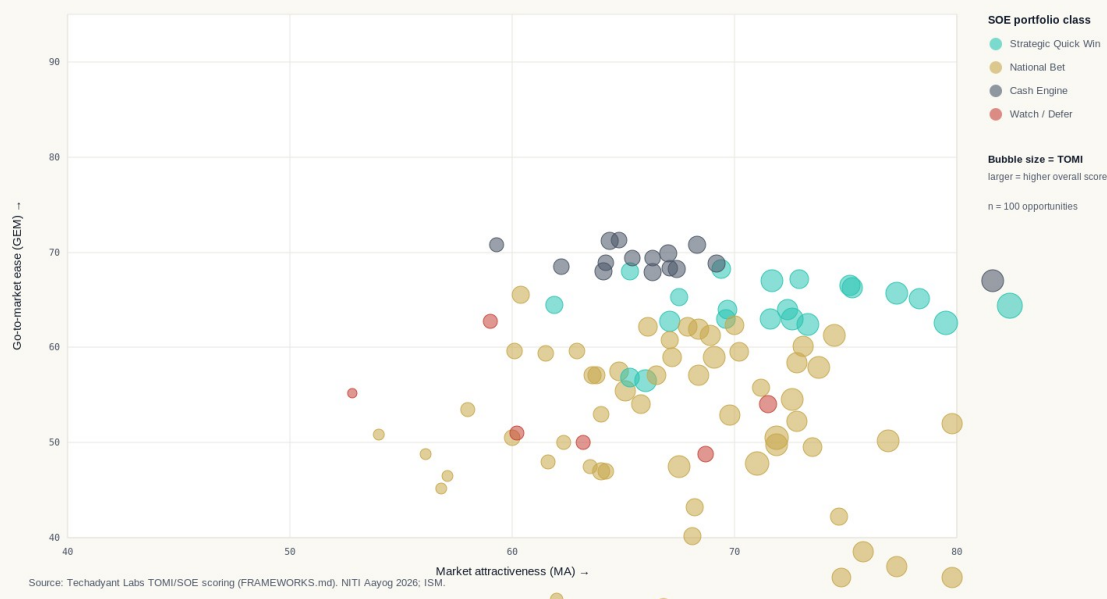


Figure 2 — The top-100 opportunity landscape (market attractiveness vs go-to-market ease, by portfolio class).

The top ten opportunities

Table 1 — The ten highest-scoring opportunities (of 100), by TOMI.

Rank	Opportunity	Part	TOMI	SOE class
1	Engineering R&D (ER&D) services export	VII	70.4	Strategic Quick Win
2	Chip-design services export (VLSI/embedded)	VII	70.0	Strategic Quick Win
3	Precision machining: vacuum chambers and UHV components	II	69.0	National Bet
4	RISC-V IP cores and design services	V	68.4	National Bet
5	Cross-border IP licensing / fabless	VII	68.3	National Bet

Rank	Opportunity	Part	TOMI	SOE class
	export			
6	OT/ICS security platform	VI	68.1	National Bet
7	AI/ML services export	VII	67.7	Cash Engine
8	AI visual quality inspection	IV	67.6	Strategic Quick Win
9	Industrial ERP for MSMEs	III	67.5	Strategic Quick Win
10	AI for semiconductor design (EDA-AI)	IV	67.5	National Bet

The single most useful finding sits in this table: the strongest opportunities are not the glamorous ones. The two highest scorers are engineering-R&D and chip-design services export — capabilities India already has at world scale, needing only the shift from being someone else's back office to owning the relationship and the IP. And no opportunity scores Tier-1: the highest TOMI is 70, firmly Tier-2. This is deliberate. Every opportunity requires a trade; the report does not pretend otherwise.

The whole map, on one page



Figure 3 — The India Technology Sovereignty Opportunity Map — where to build, and what.

What's in the full report

The paid edition (₹6,999) contains the complete analysis this free edition only sketches:

- ****All 100 opportunities, fully scored**** — one-page scorecards for every one, with attributes, sub-scores, portfolio class, investor archetype and indicative market size.
- ****Eleven parts**** mapping the opportunity layer by layer: semiconductors, industrial software, AI, engineering software, cybersecurity, and export.
- ****Ten state dashboards**** answering the question every founder has — **given what I want to build, where should I build it?**
- ****The Investment and Startup Playbooks**** — matching capital type to opportunity class, and a founder's decision tree from capability to scale.
- ****43 figures, the five proprietary frameworks in full, and the 2035 outlook.****

The full report is available at labs.techadyant.com. This free edition may be shared freely.

List of Figures

List of Tables